Teamwork Agreement

MathWorks Classroom Challenge Projects

Project: Drone Payload Capacity and Structural Design Analysis

Team Name: DronePayload2

Instructor: Kevin Graham

Program/Class: Engineering Pathway Program 2026

Members and Roles

1. Alexander Ang: MatLAB (Project Manager)
2. Julian Sanchez: CAD (Validation Lead)
3. Teppei Yoshikawa: MatLAB (Modeling Lead)
4. Stefany Chuchuca:CAD (Quality Assurance Lead)

1) Purpose & Scope

We agree to collaborate to deliver a high-quality, ethical, and reproducible project. Our work will follow the project guidance and produce the specified artifacts (e.g. plots, code, an in d visualizations).

2) Roles & Responsibilities

- Project Manager (PM): Schedules meetings; maintains timeline; ensures tasks are tracked and disagreements are resolved.
- Modeling Lead: Owns core MATLAB models/simulations
- Analysis/Validation Lead: Performs hand calculations and cross-checks simulation outputs

- Documentation & Visualization Lead: Compiles report (with figures/tables), prepares slides (if needed), and builds visual artifacts (e.g. animations or plots)
- Quality Assurance & Reproducibility Lead: Enforces coherent style across artifacts, ensures work is not overwritten across different versions, ensures data quality, tests/re-runs codes to ensure robustness and reproducibility

3) Communication Norms

- Primary channel: Discord ________________________ (e.g. Slack)

- Secondary channel: Email _______________________ (e.g. E-mail for summaries/decisions)

- Response time: ______Within 24 hours on weekdays and __________________________ on weekends (e.g. within 24 hours on weekdays; within 48 hours on weekends)

- Meeting cadence: _Once a week for an hour on the weekend__________________________ (e.g. weekly working meeting (60–90 minutes on Tuesdays via Zoom)

- Decision records: _Google Docs_____________________ (e.g. PM logs major decisions in a running document)

4) Collaboration & Tools

- Version control: ______Github Repository___________________________ (e.g. one shared GitHub repo or Google Drive folder)
- Data management: All datasets, parameters, and assumptions documented in

  _________README.md__________________ (e.g. README.md)

File naming & style: ______________Snake_case_________________ (e.g. Snake_case, descriptive names)

- Backups: _____________GIT________________________ (e.g. PM tags old versions according to numbered system)

5) Quality Standards (Definition of 'Done')

runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report

e.g. Requirements meet project description, code is reproducible (runs from the README.md file, all specified figures regenerate, at least one teammate has tested edge cases, all assumptions, units, and references are documented in report)

6) Timeline & Milestones

Week 1: Problem framing, literature/background review, dataset/parameters

Week 2: First pass models (hand calculations + baseline MATLAB scripts); initial plots/visuals.

Week 3: Validation and quality assurance

Week 4: Report, slides, presentation rehearsal, repository cleanup)

7) Decision-Making & Conflict Resolution

- Default: ________________Vote/Majority Agreement_________________________ (e.g. consensus after discussion; PM facilitates
- If blocked >48 hrs: ____________Vote/Majority Agreement___________________ (e.g. Time-limited debate, majority vote, PM documents decision)

- Technical disagreements: ________Voice call to discuss___________________
  (e.g. Evidence-based comparison before vote)

- Escalation: If unresolved, consult instructor/mentor with a concise memo.

8) Workload, Accountability & Attendance

- Task board: To-Do / Doing / Review / Done with owners and due dates documented in README.md

- Accountability: Missed deadline → recovery plan within 24 hrs; repeated misses trigger role redistribution.

- Attendance: If you'll miss a meeting, give 24-hour notice and share updates asynchronously.

10) Accessibility & Inclusion

We commit to respectful collaboration, equitable turn-taking, and flexibility for documented accessibility needs.

11) Deliverables Checklist

- [ ] Sketch out design of drone arms and choose two main designs

- [ ] CAD designs using Fusion 360

- [ ] Write out code for FEA

  Make final report showing results

- [ ] (e.g. Code repository with clear structure and README, hand calculations linked to modeled results for cross-validation, specific figures & visualizations, final report with specified sections)

12) Agreement & Signatures

Name & Signature: Alexander Ang Date: 07/09/26

Name & Signature: Teppei Yoshikawa Date: 07/09/26

Name & Signature: Julian Sanchez Date: 07/09/26

Name & Signature: Stefany Chuchuca Date: 07/09/26